

## EXPERIMENT:11

Linear Search ▾

### LINEAR SEARCH

Create a C program to perform Linear Search on an array of 10 integers. The program should allow manual input or random generation of array elements. It must display the first position of the

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],i,key,pos = -1,count = 0;
5
6     printf("enter 10 integers:\n");
7     for(i=0;i<10;i++)
8         scanf("%d",&a[i]);
9
10    printf("enter target value:");
11    scanf("%d",&key);
12
13    for(i=0;i<10;i++)
14    {
15        count++;
16        if(a[i] == key)
17        {
18            pos = i+1;
19            break;
```

45 23 67 12 89 34 56  
78 90 11 34

```
27 }
28     else
29     {
30         printf("target %d not found\n",key);
31         printf("comparisons made:%d\n",count);
32         printf("stats:Not found\n");
33     }
34     return 0;
35
36
37 }
```

### OUTPUT

```
enter 10 integers:
enter target value:target 34 found at position 6
comparisons made:6
status:Found
```

## EXPERIMENT:12

Binary Search

## BINARY SEARCH

"Write a C program implementing binary search on a sorted array of 15 integers. The program must: (1) accept an unsorted array and sort it first, (2) perform binary search, (3) display search

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[15],i,j,temp,key;
5     int low,high,mid;
6
7     printf("enter 15 number:\n");
8     for(i=0;i<15;i++)
9         scanf("%d",&a[i]);
10
11     for(i=0;i<14;i++)
12         for(j=i+1;j<15;j++)
13             if(a[i]>a[j])
14             {
15                 temp=a[i];
16                 a[i]=a[j];
17                 a[j]=temp;
18             }
19     printf("sorted array:\n");
```

```
50 30 10 90 40 70 20
80 60 5 45 35 25 55
15
45
```

```
38     else if(key>a[mid])
39         low=mid+1;
40     else
41         high=mid-1;
42 }
43 printf("element not found");
44
45 return 0;
46 }
```

## OUTPUT

```
enter 15 number:
sorted array:
51015202530354045505560708090
Enter target:element found at position 9
```

## EXPERIMENT:13

## BINARY SEARCH-ITERATIVE

Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day  $\geq 30^\circ\text{C}$  and last day  $\leq 25^\circ\text{C}$ . Display temperatures, indices (0-based), or

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[50],n,i,j,temp;
5     int first = -1,last = -1;
6
7     printf("Enter n:");
8     scanf("%d",&n);
9
10    printf("enter temperatures:\n");
11    for(i=0;i<n;i++)
12        scanf("%d",&a[i]);
13
14    for(i=0;i<n-1;i++)
15        for(j=i+1;j<n;j++)
16            if(a[i]>a[j])
17            {
18                temp=a[i];
19                a[i]=a[j];
20                a[j]=temp;
```

6

22 28 31 24 29 32

```
45     else
46         printf("\nFirst >= 30:Not found");
47
48     if(last != -1)
49         printf("\nLast <= 25: index %d(%d)",last,a[last]);
50     else
51         printf("\nLast <= 25: Not found");
52
53     return 0;
54 }
```

## OUTPUT

```
Enter n:enter temperatures:
sorted array:
222428293132
First>=30:index 4(31)
Last <= 25: index 1(24)
```

## EXPERIMENT:14

Linear Search -Sort ▾

## LINEAR SEARCH - SORT

Read n book page counts into an array. Sort ascending using any method. Use linear search to find the first book having  $\geq 300$  pages and the last book having  $\leq 150$  pages. Display page counts and indices (0-

```
1 #include<stdio.h>
2 int main(){
3     int a[50],n,i,j,temp;
4     int first = -1,last = -1;
5
6     printf("enter n:");
7     scanf("%d",&n);
8
9     printf("enter page counts:\n");
10    for(i=0;i<n;i++)
11        scanf("%d",&a[i]);
12
13    for(i=0;i<n-1;i++)
14        for(j=i+1;j<n;j++)
15            if(a[i]>a[j])
16            {
17                temp=a[i];
18                a[i]=a[j];
19                a[j]=temp;
20    }
```

7  
120 450 275 310 140  
500 200

```
45
46     if(last != -1)
47         printf("\nLast <= 150: index %d(%d)",last,a[last]);
48     else
49         printf("\nLast <= 150;not found");
50
51     return 0;
52 }
```

### OUTPUT

```
enter n:enter page counts:
sorted array:
120140200275310450500
First >= 300:index 4(310)
Last <= 150: index 1(140)
```

## EXPERIMENT:15

Linear Search - Ana ▾

## LINEAR SEARCH - ANAGRAMS

"Read n strings, use linear search to find all anagrams of a target string (same chars, any order). Display positions (1-based) and matching strings. Case-insensitive; n<=50. Sample Input: 5

```
1 #include<stdio.h>
2 #include<string.h>
3 #include<ctype.h>
4
5 void sort(char str[]){
6     int i,j;
7     char temp;
8     int n=strlen(str);
9
10    for(i=0;i<n-1;i++){
11        for(j=i+1;j<n;j++){
12            if(tolower(str[i])>tolower(str[j])){
13                temp=str[i];
14                str[i]=str[j];
15                str[j]=temp;
16            }
17        }
18    }
19 }
```

5  
listen  
race  
hello  
abc  
xyz  
slient

```
57 }
58 if(found==0)
59     printf("No anagrams found");
60
61 return 0;
62 }
```

### OUTPUT

Security Error: Disallowed code pattern detected.

## EXPERIMENT:16

Linear Search on Array

## LINEAR SEARCH ON ARRAY

"Develop a C program to perform Linear Search on an array of N integers. Accept array size dynamically

Sample Input/Output

Enter size: 5

Array: 10 20 30 40 50

Enter target: 40

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,key,pos=-1,count=0;
5     scanf("%d",&n);
6
7     int a[n];
8
9     for(i=0;i<n;i++)
10         scanf("%d",&a[i]);
11
12     scanf("%d",&key);
13
14     printf("comparison trace:\n");
15     for(i=0;i<n;i++)
16     {
17         count++;
18         if(a[i]==key)
19         {
20             printf("%d=%d\n",a[i],key);
```

```
5
10 20 30 40 50
40
```

```
33     }
34     else
35     {
36         printf("target Not found\n");
37         printf("comparison made = %d\n",count);
38     }
39     return 0;
40 }
```

## OUTPUT

```
comparison trace:
10 != 40
20 != 40
30 != 40
40=40
target found at position 4
comparison made = 4
```

## EXPERIMENT:17

Binary Search - Arr. ▾

## BINARY SEARCH - ARRAY

Write a C program to perform Binary Search on a sorted array of 10 integers. The program should accept the array elements and a target value from the user. It must display the position of the element and the total

```
1 #include<stdio.h>
2
3 int main()
4 {
5     int a[10],i,key;
6     int low=0,high=9,mid;
7     int count=0,found=0;
8
9     printf("enter 10 sorted integers:\n");
10    for(i=0;i<10;i++)
11        scanf("%d",&a[i]);
12
13    printf("enter element to search:");
14    scanf("%d",&key);
15
16    while(low<=high)
17    {
18        count++;
19        mid=(low+high)/2;
```

5 12 18 23 34 45 56

67 78 90

45

```
35    }
36 }
37 if(found==0){
38     printf("element not found\n");
39     printf("comparisons made:%d\n",count);
40 }
41 return 0;
42 }
```

## OUTPUT

```
enter 10 sorted integers:
enter element to search:element 45 found at position 6
comparisons made:3
```

## EXPERIMENT:18

Linear Search - Tem

## LINEAR SEARCH - TEMPERATURE

"Read n daily temperatures into an array. Use Linear Search to find the last day whose temperature is less than 20°C. Display temperatures, index (0-based), and value. If not found. print ""Not

```
1 #include<stdio.h>
2
3 int main()
4 {
5     int n,i,a[50],index = -1;
6     scanf("%d",&n);
7
8     if(n > 50)
9     {
10        printf("invalid");
11        return 0;
12    }
13    for(i = 0; i < n; i++)
14        scanf("%d",&a[i]);
15
16    for(i = 0; i < n; i++)
17        printf("%d",a[i]);
18
19    for(i = 0; i < n; i++)
20    {
```

7  
18 25 21 19 30 17 24

```
23     }
24     if(index == -1)
25         printf("\nNot found");
26     else
27         printf("\nLast <20: index %d(%d)",index,a[index]);
28
29     return 0;
30 }
```

## OUTPUT

```
18252119301724
Last <20: index 5(17)
```

EXPERIMENT:19



Linear Search -Price ▾

## LINEAR SEARCH - PRICES

"Read n product prices into an array. Sort ascending using any method. Use linear search to find the first price  $\geq ₹500$  and the last price  $\leq ₹200$ . Display prices and indices (0-based), or "Not found". Validate ▾

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,a[50],i,j,t;
5     int first = -1,last = -1;
6
7     scanf("%d",&n);
8     for(i=0;i<n;i++)
9         scanf("%d",&a[i]);
10
11     for(i=0;i<n-1;i++)
12         for(j=i+1;j<n;j++)
13             if(a[i]>a[j])
14             {
15                 t = a[i];
16                 a[i] = a[j];
17                 a[j] = t;
18             }
19     for(i=0;i<n;i++)
20         printf("%d ",a[i]);
```

7  
150 650 220 90 500  
780 180

```
28     for(i=n-1;i>=0;i--)
29         if(a[i]<=200)
30         {
31             last=i;
32             break;
33         }
34     printf("\nfirst >= 500:index %d(%d)",first,a[first]);
35     printf("\nLast <= 200:index %d(%d)",last,a[last]);
36
37     return 0;
38 }
```

## OUTPUT

```
90150180220500650780
first >= 500:index 4(500)
Last <= 200:index 2(180)
```

## EXPERIMENT:20

Bin. Search-Iterativ ▾

### BIN. SEARCH-ITERATIVE-TEMP

"Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day  $\geq 30^{\circ}\text{C}$  and last day  $\leq 25^{\circ}\text{C}$ . Display temperatures, indices (0-based). or

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,a[50],i,j,t;
5     int first = -1,last= -1;
6
7     scanf("%d",&n);
8
9     for(i=0;i<n;i++)
10         scanf("%d",&a[i]);
11
12     for(i=0;i<n-1;i++)
13         for(j=1+1;j<n;j++)
14             if(a[i]>a[j])
15             {
16                 t=a[i];
17                 a[i]=a[j];
18                 a[j]=t;
19     }
```

6  
22 28 31 24 29 32

```
29 {
30     if(a[i]<=25)
31     {
32         last=i;
33         break;
34     }
35 }
36 printf("\nFirst >= 30: index %d(%d)",first,a[first]);
37 printf("\nLast <=25: index %d(%d)",last,a[last]);
38
39 return 0;
40 }
```

### OUTPUT

```
First >= 30: index 2(31)
Last <=25: index 1(24)
```